Machine Learning Project

# Customer Churn
# Prediction App

Predicting customer retention based on behavioral and subscription data using a high-performance XGBoost pipeline.

Streamlit UI

XGBoost Engine

# 🚀 Project Overview

Identifying customers likely to stop using a service to enable proactive retention strategies.

### Supervised Learning

Building a robust classification pipeline to categorize customers into two distinct states.

### Classification Labels

→ **0** Not Churn (Retained)

→ **1** Churn (Exited)

**Deployment:** Final model serves real-time predictions via an interactive Streamlit web application.

# 🧠 Machine Learning Pipeline

### Preprocessing

→ Numerical: **StandardScaler**

→ Categorical: **OneHotEncoder**

### Final Selection

## XGBoost

Chosen for superior accuracy and handling of non-linear relationships.

Other models tested: Logistic Regression, Random Forest, and SVM.

# 📊 Features Used

### Numerical Attributes

Age, Tenure, Usage Frequency, Support Calls, Payment Delay, Total Spend, Last Interaction.

### Categorical Attributes

Gender, Subscription Type (Basic, Standard, Premium), Contract Length.

### Data Quality

Features were selected based on their predictive power regarding customer subscription lifecycles.

# ⚙️ Tech Stack

- Python
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Streamlit
- Joblib

# 🖥️ Streamlit App Features

- → Manual customer input form for testing.
- → Real-time churn prediction engine.
- → Probability score output for risk assessment.
- → Pre-trained model loading (churn_model.pkl).
- → Fast inference with zero retraining lag.

**Ready for Production**